HumorRank: A Tournament-Based Leaderboard for Evaluating Humor Generation in Large Language Models

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Abstract

Evaluating humor in large language models (LLMs) is an open challenge because existing approaches yield isolated, incomparable metrics rather than unified model rankings, making it difficult to track progress across systems. We introduce HumorRank, a tournament-based evaluation framework and leaderboard for textual humor generation. Using SemEval-2026 MWAHAHA test dataset, we conduct an extensive automated pairwise evaluation across nine models spanning proprietary, open-weight, and specialized systems. Pairwise judgments grounded in the General Theory of Verbal Humor (GTVH) are aggregated via an Adaptive Swiss tournament, with Bradley-Terry Maximum Likelihood Estimation (MLE) producing globally consistent capability rankings. Our results demonstrate that HumorRank yields statistically grounded model stratifications, showing that humor quality is driven by mastery of comedic mechanisms rather than model scale alone. HumorRank thus provides a scalable, interpretable methodology for benchmarking and understanding LLM-generated humor.

1 Introduction

Humor generation is a domain that requires a highly nuanced understanding of language, context, and pragmatic reasoning (Quan et al., 2025; Kim & Chilton, 2025), posing a significant challenge for evaluating the capabilities of large language models (LLMs) (Narad et al., 2025). This difficulty is reflected in the fragmented landscape of existing evaluation methods, where different works adopt incompatible paradigms, including humor detection (Ajayi & Mitra, 2025b; Romanowski et al., 2025), scalar scoring (Goes et al., 2022), classification (Wu et al., 2025a), LLM-as-a-Judge approaches (Shafiei & Saffari, 2025), and costly human preference evaluations (Romanowski et al., 2025; Horvitz et al., 2024).

A central limitation of these approaches is the lack of a unified and scalable framework for comparing models. Existing methods measure different aspects of humor and do not produce consistent rankings across systems, making it difficult to track progress. As LLMs increasingly operate in interactive and creative settings, establishing a reliable and comparable evaluation protocol for humor generation becomes essential.

To address this gap, we introduce **HumorRank**, a leaderboard-oriented framework for ranking humor generation in language models. HumorRank casts evaluation as a pairwise preference problem and aggregates outcomes into a globally consistent ranking. We evaluate nine models on the SemEval-2026 Task 1: MWAHAHA dataset (Castro et al., 2026), demonstrating scalable, interpretable, and reproducible comparison of humor generation systems.

Our contributions are as follows:

1. We introduce **HumorRank**, a scalable framework that formulates humor evaluation as a global ranking problem over competing language models.

2. We formalize humor assessment as a **pairwise preference learning task** and show that Bradley–Terry estimation yields stable and comparable rankings.
3. We develop a **theory-grounded LLM-as-Judge** protocol that produces structured and **interpretable** evaluation signals, and demonstrate its effectiveness through a large-scale comparative study on SemEval-2026 MWAHAHA with consistent cross-judge rankings.

2 HumorRank

The subjective and multidimensional nature of humor presents fundamental challenges to absolute quality scoring. To address this, we operationalize humor theoretically as a continuous *cognitive reward* arising from the successful resolution of deliberately constructed linguistic incongruities (for our full formal definition and its theoretical derivation, see Appendix A). Because lexical and semantic features such as comedic delivery (Romanowski et al., 2025; Kim & Chilton, 2025) interact in ways that resist direct quantification (Winters & Van der Stockt, 2025), pairwise comparison mitigates these limitationsRavi et al. (2024) by constraining the evaluation task to a relative preference judgment between two model-generated jokes conditioned on the same prompt (Hossain et al., 2020). This formulation substantially reduces the cognitive load on the evaluator and is more robust to inter-annotator variance than uncalibrated scalar annotation.

While pairwise comparisons provide high-fidelity local signal, they remain fundamentally discrete and unordered—insufficient, in isolation, to support a system-level leaderboard. To transform a collection of $K2$ pairwise outcomes over K competing models into a globally consistent capability ranking, an aggregation framework capable of resolving local inconsistencies and propagating information across the full tournament graph is required.

HumorRank addresses this through a two-stage pipeline: an Adaptive Swiss Tournament that efficiently generates the pairwise comparison graph, and a Bradley-Terry(BT) global Maximum Likelihood Estimation (MLE) that maps the observed outcomes to statistically grounded, continuous capability estimates. We additionally derive Stable Elo ratings as a secondary reference metric, reported alongside BT scores for cross-validation purposes.

2.1 Bradley-Terry Global Maximum Likelihood Estimation

The Bradley-Terry model (Bradley & Terry, 1952) serves as our primary, order-independent ranking algorithm in this work. By maximizing the likelihood of the observed pairwise outcomes across the entire tournament graph, the BT model estimates a latent "humor capability" score for each model.

Given models i and j with latent ratings R_i and R_j, the probability that model i wins over model j is formulated as an Elo-scaled logistic function:

$$P(iwinsagainstj) = \frac{1}{1 + 10^{(R_j - R_i)/400}} \tag{1}$$

Instead of sequential updates, HumorRank fits the global Maximum Likelihood Estimation (MLE) using the iterative Minorization-Maximization (MM) algorithm until a strict convergence tolerance ($\epsilon < 10^{-6}$) is met and ratings are anchored at a base of 1000. To establish statistical significance and prove that the separation between model tiers is rigorous, we compute and report 95% confidence intervals by resampling the match history with replacement via 100 bootstrap iterations.

2.2 Stable Elo (Sequential Reference)

While the BT model provides the global MLE, we simultaneously compute a sequential Elo ratingAlbers & Vries (2001) to track dynamic stability and provide a secondary reference metric. The generalized sequential update rule is:

$$R_{new} = R_{curr} + K_{fac} \cdot (S - E) \tag{2}$$

where $K_{fac} = 32$ specifies the maximum volatility factor, S denotes the observed outcome (1.0 for a win, 0.5 for a tie, 0.0 for a loss), and E is the expected probability derived from Equation 1.

A known deficiency of standard Elo is order dependence, wherein the specific sequence of matches heavily influences the final ratings. HumorRank mitigates this vulnerability by implementing *Stable Elo*: the entire tournament history is replayed across $N = 5$ randomly shuffled topological orderings. The final assigned score is the arithmetic mean of the resulting terminal ratings, ensuring sequence-agnostic stability (see Appendix I for convergence proofs across shuffles).

2.3 Adaptive Swiss Pairing

For large model pools, exhaustive $\mathcal{O}(K^2)$ pairwise comparisons become financially and computationally prohibitive. The HumorRank tournament engine resolves this through an *Adaptive Swiss Pairing* algorithm controlled by a single budget parameter $C_{\max}$. Rather than random assignment, the engine preferentially matches models of similar standing while avoiding previously observed pairings, maximizing information gain per match and driving Bradley–Terry convergence in $\mathcal{O}(K \log K)$ comparisons. The coverage depth is determined entirely by $C_{\max}$: setting $C_{\max} = K2 \times P$ (where P denotes the number of prompts) yields exhaustive pair coverage across all prompts; reducing $C_{\max}$ below this threshold recovers the sublinear efficiency without altering the pairing logic (see Algorithm 1).

Algorithm 1 Adaptive Swiss Pairing for Tournament Evaluation

Require: Set of models $\mathcal{M}$, max comparisons C_{max}, tracking Elo ratings R
Initialize match history graph $G \leftarrow (V = \mathcal{M}, E = \emptyset)$
while $|E| < C_{max}$ **do**
  Sort $\mathcal{M}$ descending by temporal Elo ratings R
  Initialize unmatched subset $U \leftarrow \mathcal{M}$
  Initialize round pairings $P \leftarrow \emptyset$
  while $|U| \geq 2$ **do**
    $i \leftarrow$ highest-rated model in U
    Find $j \in U \setminus \{i\}$ minimizing $|R_i - R_j|$ where (i, j) is under-sampled in G
    if valid match j exists **then**
      $P \leftarrow P \cup \{(i, j)\}$
      $U \leftarrow U \setminus \{i, j\}$
    else
      break ▷ Fallback resolution for pairing exhaustion
    end if
  end while
  Execute LLM-as-a-judge evaluations for all pairs in P
  Update match edges $E \leftarrow E \cup P$
  Update tracking ratings R using observed outcomes
end while
return Global match history graph G

3 Experimental Setup

To empirically validate the HumorRank methodology, we execute a comprehensive large-scale evaluation using officially recognized humor generation benchmarks. Our experimental design explicitly tests the framework's discriminative power across varying model architectures, access paradigms, and parameter scales. Full reproducibility details, including hyperparameters and computational budget, are provided in Appendix D.

3.1 Dataset: SemEval-2026 MWAHAHA

We leverage the full test set of the SemEval-2026 Task 1: MWAHAHA dataset (Castro et al., 2026). The dataset comprises 300 distinct prompt conditions encompassing both headline-based associative humor and constrained word-combination tasks. By utilizing an established, external, and heterogeneous test set designed as part of the competiton, we ensure robust domain generalizability and inoculate the evaluation against selective cherry-picking.

3.2 Model Evaluation Suite

We evaluate a deliberately diverse suite of 9 language models to assess the leaderboard's capacity to resolve fine-grained capability differences. The inclusion criteria strictly span multiple model lineages and access paradigms:

- **Frontier Proprietary Models:** GPT-5Singh et al. (2025), Gemini 2.5 ProComanici et al. (2025), Claude 3.5 HaikuAnthropic (2024), and Kimi K2Team et al. (2026).
- **Open-Weight Competitive Models:** Qwen 3 32BBai et al. (2025), GPT OSS 120BAgarwal et al. (2025), and Llama 3.3 70B InstructGrattafiori et al. (2024).
- **Specialized and Baseline Models:** *HumorGen 7B* (Ajayi & Mitra, 2025a) (a fine-tuned specialist designed explicitly for humor generation) alongside *Base Qwen 7B*Team (2024) acting as the critical zero-shot architecture control.

3.3 Evaluation Protocol and LLM-as-Judge Ablation

HumorRank employs open-weight models as pairwise judges, executing the full round-robin schedule across all 9 contestants ($92 = 36$ matches per prompt $\times$ 300 prompts $=$ 10,800 total automated judgments). We structure the judgeship around two complementary roles.

Primary Judge: Llama 3.3 70B Instruct (quantized for local inference) serves as the primary judge for all headline rankings, Bradley-Terry derivations, and feature analyses. The exact prompt templates utilized for adjudication are reproduced in Appendix **??**.

Judge Ablation & Validity Check: A recurrent critique of LLM-as-a-Judge is the risk of judge-specific preference artifacts. To address this, we independently re-evaluate the identical 10,800 matches using Qwen 2.5 72B as an alternative judge. High cross-judge rank correlation would constitute direct evidence of a latent, model-agnostic humor ordering. Additionally, to ground the automated pipeline against human perception, we conduct a controlled annotation study on 50 representative pairwise matches, with independent annotators blind to model identity (annotator guidelines and inter-annotator agreement metrics are detailed in Appendix H).

4 Results

Our evaluation yields an extensive empirical profile of humor capability across current language models. We present the system-level Bradley-Terry (BT) leaderboard, validate its stability across independent LLM judges, and subsequently decompose these ratings into interpretable psychometric features.

4.1 HumorRank Leaderboard

The main tournament, evaluated by Llama 3.3 70B across 10,800 matches, reveals a clear, statistically significant stratification of model capabilities. Table 1 details the derived BT ratings, 95% Confidence Intervals, and win rates for the evaluated suite.

Remarkably, the specialized *HumorGen-7B* model (Rank 4, BT = 1092.8) successfully bridges the gap between the mid-tier open-weights and the proprietary frontier, cleanly outperforming models an order of magnitude larger (e.g., GPT OSS 120B, Rank 6) as shown in Table 1

Rank	Model	BT Rating	95% CI	Win Rate
1	GPT-5	1307.5	[1288.5, 1325.9]	84.0%
2	Kimi-K2	1156.9	[1139.9, 1169.1]	67.8%
3	Gemini 2.5 Pro	1115.1	[1096.6, 1128.1]	62.6%
4	**HumorGen SFT 7B**	**1092.8**	**[1077.6, 1108.5]**	**59.8%**
5	Claude 3.5 Haiku	1037.5	[1022.4, 1050.9]	52.7%
6	GPT OSS 120B	1015.0	[1004.2, 1029.9]	49.8%
7	Qwen 3 32B	976.9	[965.0, 987.8]	45.0%
8	Llama 3.3 70B	761.0	[745.1, 781.4]	21.8%
9	Base Qwen 7B	537.4	[513.5, 566.7]	6.5%

Table 1: Primary HumorRank Leaderboard (Llama 3.3 70B Judge). Models cleanly separate into a Frontier Elite tier (> 1100), a Competitive Mid-Tier ($970 - 1100$), and a Weak Baseline (< 800).

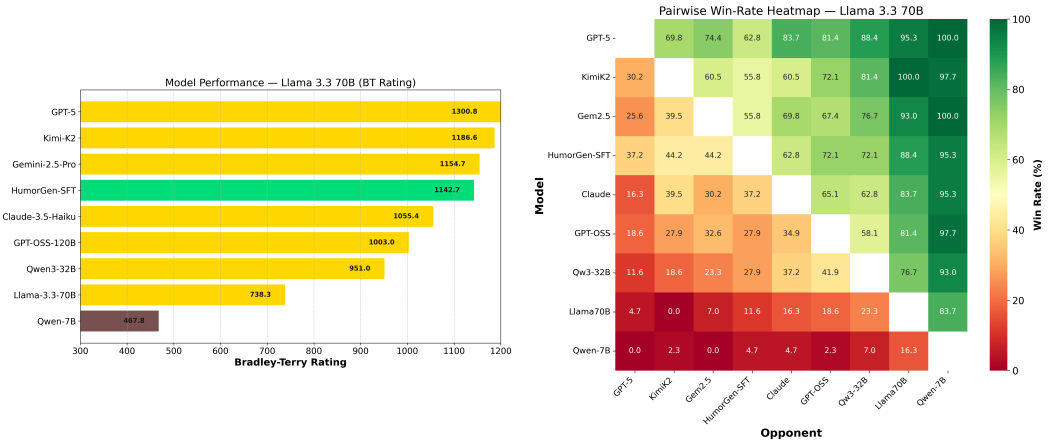


Figure 1: HumorRank Leaderboard (left) and Pairwise Win-Rate Heatmap (right) showing the performance of the 9 models.

and Figure 1. It is also notable that the Llama 3.3 70B judge penalizes its own generations (ranking itself 8th out of 9, BT = 761.0; Table 1), providing a strong empirical counter-signal to frequent concerns regarding LLM self-preference bias.

4.2 Cross-Judge Validity and Rank Stability

Evaluating subjective data is inherently sensitive to the choice of the primary evaluator. To confirm that the derived rankings represent a generalized latent humor ordering rather than Llama-specific architectural artifacts, we conduct a full ablation running the identical 10,800 tournament matches utilizing the Qwen 2.5 72B evaluator with details of the ranking shown in the Appendix B.

The resulting Qwen-adjudicated BT ratings exhibit exceptionally high rank correlation against the primary Leaderboard, achieving a **Kendall's** $\tau = 0.889$ **(**$p < 0.01$**)**. The top-two models (GPT-5, Kimi K2) and the bottom-three models remain rigidly fixed. Mid-tier variance is minimal; notably, HumorGen SFT 7B shifts to #3 under the Qwen judge (Appendix B), suggesting its underlying humor quality is robust to evaluator identity. The absence of intransitive loops (Transitivity Score = 1.0) across both paradigms validates the use of BT modeling for this domain.

4.3 Human Evaluation

To assess agreement between human judgments and the automated evaluator, we conducted a blind annotation study using our pairwise comparison protocol. Two independent expert evaluators reviewed anonymized outputs across 60 randomly sampled pairs, selecting the funnier response or indicating a tie.

Our evaluation (120 total votes across 2 evaluators) yields a Human Agreement of 50.0%. Inter-annotator reliability, measured via Krippendorff's Alpha for nominal data, is $\alpha = 0.432$ (moderate agreement), providing a reliable benchmark despite humor's subjectivity. Comparing human consensus to the primary Llama 3.3 70B judge, Gold Standard alignment (full consensus matches) is 53.3%, while overall LLM Agreement (micro-averaged accuracy per vote) is 51.7%. Notably, LLM Agreement slightly exceeds human-to-human agreement, indicating closer alignment with the emerging consensus than between annotators.

4.4 Theory-Grounded Feature Interpretability

To transcend zero-dimensional scalar leaderboards, HumorRank utilizes a Layered Psychometric Model grounded in the General Theory of Verbal Humor (GTVH) (Attardo, 2017), detailed in Table 2. We isolate *Humor Mechanisms* (deep semantic incongruities), *Delivery Features* (surface psycholinguistic presentations), and *Failure Modes*.

Feature	GTVH Resource	Theoretical Description
Incongruity	Logical Mechanism (LM)	The foundational juxtaposition of conflicting scripts or ideas, forming the core of structural humor.
Absurdity	Situation (SI)	Extreme logical divergence from reality; scenarios that break fundamental physical or social expectations.
Sarcasm	Target (TA) / LM	Irony deployed with a specific target; relies on the contrast between stated and intended meaning.
Wordplay	Language (LA)	Surface-level phonetic or lexical ambiguity, including puns, double entendres, and syntactic misdirection.
Conciseness	Language (LA) / Delivery	The efficiency of the buildup before the punchline; minimizing superfluous tokens to maximize comedic timing.

Table 2: Taxonomy of Humor Features extracted from model generations, grounded in the General Theory of Verbal Humor (GTVH).

Analysis of automated tagging over 10,000 wins delineates highly distinctive comedic signatures across three model archetypes:

- **The Frontier Generalist (e.g., GPT-5):** Relies heavily on pristine delivery, exhibiting exceptionally high *Conciseness* (over 30% of wins) alongside standard *Incongruity* (Figure 2, right). Its polished, safe generations result in *Cliché* as its primary failure mode.
- **The Absurdist Specialist (e.g., HumorGen 7B):** Surpasses its scale limitation by mastering deep structural mechanisms—the highest corpus rates of *Absurdity* (25.8%) and *Sarcasm* (9.2%) across all models (Figure 2, left)—leaning on *Escalation* rather than brevity (Figure 2, right).
- **The Weak Baseline (e.g., Llama 70B):** Over-indexes on superficial *Wordplay* (29.5%; Figure 2, left) while chronically suffering from *Weak Punchlines* (45% of failure tags; Table 3), lacking the structural commitment of higher-rated models.

Table 3 presents a concrete pairwise example illustrating these distinctions. The corresponding per-model feature distributions, adjudicated by the primary Llama 3.3 70B judge, are visualized in Figure 2; Qwen 2.5 72B results are provided in Appendix F for cross-judge comparison.

Model	Generated Output	GTVH Tags & Reasoning
HumorGen 7B *(Winner)*	"My doctor told me to watch my drinking, so I bought a mirror for the bar."	**Absurdity, Conciseness.** Rapid escalation into an absurd literal interpretation of a medical idiom; punchline delivered with zero wasted context.
Llama 70B *(Loser)*	"Why did the doctor tell the man to watch his drinking? Because he wanted him to see the glass half full of trouble!"	**Cliché, Weak Punchline.** Overused idiom; superficial wordplay that fails to introduce genuine semantic incongruity.

Table 3: Qualitative pairwise example illustrating contrasting GTVH feature profiles of a winning vs. losing generation.

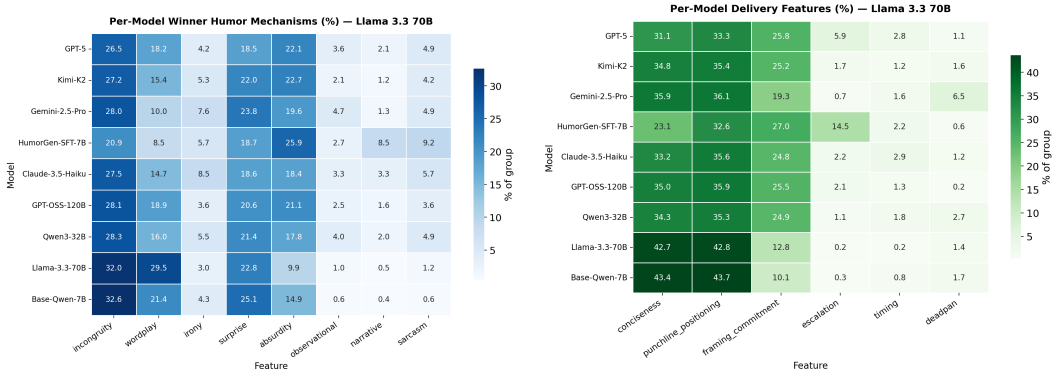


Figure 2: Per-model winning feature distributions (Llama 3.3 70B judge). **Left:** Humor mechanisms (% of wins). **Right:** Delivery features (% of wins). Frontier models dominate via *Conciseness*; the specialist model leads on *Absurdity* and *Escalation*; baseline models over-index on *Wordplay*.

Complementing the winner profiles, Figure 3 reveals the failure mode distribution for each model. Notably, the dominant failure modes for most models are *Cliché* and *Weak Punchline*—but *HumorGen SFT 7B* exhibits a strikingly different signature: it accrues substantially higher *Overexplained* (25.2%) and *Buried Punchline* (20.4%) failure rates than any other model, indicating that its aggressive structural depth occasionally overreaches—building setups that collapse before delivery rather than landing superficial wordplay.

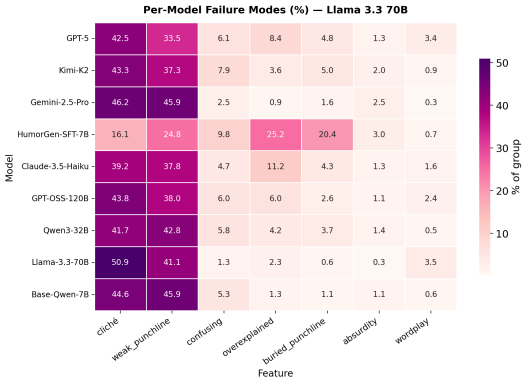


Figure 3: Per-model failure mode distributions (Llama 3.3 70B judge). *Cliché* and *Weak Punchline* dominate most models, but HumorGen SFT 7B stands out with markedly higher *Overexplained* and *Buried Punchline* rates—a byproduct of its deep-structure comedic strategy. Qwen 2.5 72B failure modes are in Appendix F.

This feature landscape confirms that scaling alone yields refined delivery, but specialized alignment is necessary for deep structural absurdity—a finding consistent across both judge architectures (Appendix F).

5 Related Works

5.1 Model Evaluation in Humor Generation Systems

Despite growing interest in LLM humor capabilities, existing evaluation protocols remain inconsistent and largely incomparable across studies. Existing approaches span across a wide range of paradigms: automated metrics applied to human-AI co-creative humor Wu et al. (2025b); crowd-sourced AI voting panels that assess different humor types independently Goes et al. (2022); Best-Worst Scaling (BWS), which elicits relative judgments across joke sets rather than absolute ratings Yamane (2024); funniness rating scales and Likert-style templates Gorenz & Schwarz (2024); and fully human evaluation, which—while considered the gold standard—is costly, slow, and typically limited to small validation sets Zhang et al. (2024); Goel et al. (2024); Wang et al. (2025); Jain et al. (2024). Broader evaluations of LLM humor understanding and generation ability Ajayi & Mitra (2025b); Zhou et al. (2025); Song et al. (2025) similarly rely on these fragmented protocols.

Critically, none of these approaches produce *ranked preference orderings* across multiple generation systems, nor do they provide interpretable rationale for their judgments. Each evaluation is essentially a one-off measurement rather than a scalable, comparative framework. This limitation stands in contrast to the broader NLP evaluation literature, where tournament-based rating systems have emerged as a principled and scalable alternative—a development we discuss in Section 5.3.

5.2 Computational Humor: Datasets, Theory, and Generation

The study of humor is rooted in psychology Larkin-Galiñanes (2017) and linguistics Attardo (2024), with classical theories such as superiority, relief, and incongruity providing explanatory accounts of what constitutes humor Veatch (1998). These frameworks motivate interpretable dimensions of humor—such as expectation violation, tension release, and social positioning—that remain influential in computational modeling. However, they do not provide a deterministic recipe for generationLarkin-Galiñanes (2017) because humor is inherently subjective, varying across context, culture, and individual perception. Despite this, linguistic and pragmatic analysis identify relatively stable cues—such as timing, delivery, ambiguity, and form–meaning incongruity—that support both dataset construction and automated evaluation.

Building on these foundations, prior work has introduced a range of humor benchmarks, historically focused on text-based tasks. More recently, advances in large language models have expanded this landscape to include multimodal datasets and evaluation settings spanning humor generation, understanding, and ranking Zhong et al. (2024); Zhang et al. (2024); He et al. (2024); Ryan et al. (2025); Jain et al. (2024). While these developments broaden the empirical scope of computational humor, they also underscore the need for scalable and comparable evaluation methodologies. In particular, reliably assessing humor requires frameworks that can accommodate subjectivity while enabling consistent comparison across models, motivating structured, ranking-based approaches to evaluation.

5.3 LLM Leaderboard Rating Systems in NLP Tasks

The use of leaderboard-based evaluation has become increasingly prevalent in NLP, providing a standardized framework for comparing model performance across tasks and benchmarks Toloka Team (2023); Chiang et al. (2024); Myrzakhan et al. (2024). Modern leaderboard platforms, such as Chatbot Arena and the Open LLM Leaderboard, often leverage *LLM-as-a-Judge* paradigms to enable scalable and automated evaluation of model outputs Zheng et al. (2023). This approach is particularly attractive as it supports both human and model-based preference judgments, allowing for flexible and cost-effective

evaluation pipelines. Furthermore, leaderboard-based systems facilitate direct comparison of models under consistent conditions, making them a practical choice for benchmarking progress in NLP Federiakin (2025); Myrzakhan et al. (2024). Prior work suggests that such ranking frameworks provide a reliable proxy for model quality and can be adapted to diverse settings, including multilingual and domain-specific evaluation scenarios Park et al. (2024); Silva et al. (2026).

6 Conclusions

We presented **HumorRank**, a leaderboard-oriented framework for comparing LLM humor generation under conditioned prompts. The pipeline combines exhaustive pairwise coverage via **Adaptive Swiss pairing** with **Bradley–Terry MLE** (primary) and **Stable Elo** as a sequence-robust reference, enabling scalable evaluation that extends to sublinear match budgets for larger model pools. On **SemEval-2026 MWAHAHA** (300 prompts) Castro et al. (2026), we rank **nine** large language models spanning proprietary and open-weight systems. We use **Llama 3.3 70B** as the primary judge and **Qwen 2.5 72B** as a secondary evaluator, observing **high rank agreement** (Kendall's $\tau = 0.889$), demonstrating that the induced rankings are stable across distinct judge architectures.

Empirically, frontier models cluster at the top of the leaderboard; however, a **domain-specialized 7B model** (*HumorGen 7B* (Ajayi & Mitra, 2025a)) achieves competitive performance with substantially larger general-purpose systems, supporting the hypothesis that humor quality depends on mechanisms beyond scale alone. Post-hoc analysis of judge rationales, grounded in the **General Theory of Verbal Humor**, reveals consistent patterns in winning responses, highlighting the importance of delivery, timing, and structured incongruity in distinguishing high-quality humor.

HumorRank is designed as a **reusable evaluation stack**, enabling consistent comparison as new models and prompts emerge. Preliminary human evaluation indicates directional alignment between automated judgments and human preferences, with a fuller annotation study ongoing. We defer a detailed discussion of limitations to Section 7 and outline directions for expanding coverage and validation in future work. We hope this framework supports treating humor generation as a **comparable, rankable** capability rather than a collection of isolated metrics.

7 Limitations

We note several limitations of the current study:

- **Monolingual Scope:** Evaluation is conducted on English data; cross-lingual and cross-cultural humor settings are not examined in this work.
- **Model Coverage:** Experiments are limited to nine models, which do not exhaust the full space of contemporary systems.
- **Dataset Scope:** Evaluation is directly anchored to the SemEval-2026 MWAHAHA dataset. While it rigorously tests generative capability within its established distribution, it does not evaluate performance across alternative interactive or multimodal humor paradigms.

8 Reproducibility Statement

To ensure full reproducibility of the HumorRank framework, we detail the core hyperparameter configurations and computational hardware requirements necessary to execute the generative tournament.

Generation Hyperparameters: For all candidate models evaluated in the tournament, we standardized the generation settings to prioritize creative diversity while maintaining structural coherence. Specifically, we configured the candidate models with a standard

sampling temperature of $T = 0.7$. Default nucleus sampling (top-$p = 1.0$) and token limit thresholds were inherited directly from the respective model APIs to preserve native instructional adherence without imposing artificial truncation.

Judge Hyperparameters: The LLM judges (Llama 3.3 70B and Qwen 2.5 72B) were configured with a highly constrained sampling temperature of $T = 0.1$ alongside a maximum retry threshold of 3 (with exponential backoff) for all pairwise JSON evaluation calls. Given the substantial financial and computational cost of the expansive generative tournament, this near-deterministic setting strictly ensures that the judge maintains stability and does not yield erratic or contradictory evaluations to the same prompt upon reassessment, thereby firmly preserving the integrity of the Bradley-Terry ratings.

Computational Hardware: The large-scale algorithmic execution of the HumorRank tournament is densely computationally intensive. Orchestrating the complete 10,800-match evaluation sequence for the primary leaderboard rankings strictly required 48 hours of dedicated NVIDIA H100 (80GB) GPU compute time. This extensive hardware footprint is strictly required to identically replicate the vast comparative throughput of the tournament dynamics.

9 Ethics Statement

This work proposes a framework for the systematic evaluation and ranking of humor generation across large language models; it does not itself constitute a humor generation system. Two ethical considerations warrant explicit acknowledgment. First, humor is a culturally and contextually variable phenomenon whose boundaries with offensive or exclusionary expression are highly sensitive to audience and setting. Evaluation frameworks that rank models on comedic output implicitly surface content generated by those models, and practitioners adapting such pipelines for downstream applications bear responsibility for enforcing appropriate content-moderation constraints. Second, the validity of automated ranking is constrained by the cultural and stylistic distribution of the judge model's pretraining corpus. LLM-based evaluators trained predominantly on high-resource, Western-centric text may systematically disadvantage humor conventions from linguistically or culturally underrepresented communities—a limitation shared by the broader LLM-as-a-judge literature (Zheng et al., 2023). HumorRank should therefore be interpreted as a reproducible diagnostic benchmark rather than a definitive assessment of comedic or creative quality.

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A Humor Definition

Building upon classic Incongruity Theory, psychological frameworks (Larkin-Galiñanes, 2017), and Normative-Violation theory (Veatch, 1998), we require a rigorous working definition that can be applied to text-based evaluation. For the purposes of this research, we define humor explicitly as:

> **Humor is the cognitive reward—experienced as amusement—arising when an interlocutor successfully resolves a deliberately constructed incongruity, such as the narrative shift between a joke's setup and punchline, within a harmless and non-threatening context.**

This definition anchors psychological consensus into the practical reality of evaluating generated text. It explicitly requires four distinct components:

- **The Joke Mechanism (Setup & Punchline):** We evaluate humor not as random surprise, but as a structured linguistic narrative. The setup creates a logical expectation, and the punchline deliberately subverts it.
- **The "Cognitive Reward":** This maps to the cognitive appraisal process, describing the computational or intellectual achievement of bridging the logical gap between the setup and punchline.
- **Experienced as Amusement:** The cognitive resolution must trigger a pleasant response (mirth) rather than confusion.
- **Harmless Context:** Drawn from benign violation theory, the structural incongruity only produces amusement if it is appraised as non-threatening.

Because humor exists on a continuous spectrum determined by these mechanisms rather than as a discrete label, our methodology utilizes *pairwise preference ranking*. By prompting the judge to evaluate which generation produces a stronger cognitive reward, we effectively treat humor evaluation as a reward modeling paradigm across the multi-dimensional feature space of human amusement.

B HumorRank Leaderboard Performance with Qwen 2.5 72B LLM Judge

To validate the stability of our primary leaderboard, which was generated using Llama 3.3 70B as a judge, we replicated the evaluation using Qwen 2.5 72B as the judge. Figure 4 presents the resulting Bradley-Terry leaderboard from this independent evaluation pipeline.

Rank	Model	BT Rating	95% CI	Win Rate
1	GPT-5	1262.7	[1245.6, 1279.5]	80.7%
2	Kimi-K2	1126.2	[1114.0, 1137.7]	64.8%
3	**HumorGen SFT 7B**	**1100.9**	[**1086.1**, **1114.6**]	**61.5%**
4	Claude 3.5 Haiku	1062.1	[1045.8, 1076.8]	56.3%
5	Gemini 2.5 Pro	1058.5	[1046.3, 1071.8]	55.8%
6	GPT OSS 120B	1036.0	[1022.8, 1049.5]	52.9%
7	Qwen 3 32B	1009.0	[990.6, 1026.4]	49.3%
8	Llama 3.3 70B	746.3	[724.1, 761.0]	19.8%
9	Base Qwen 7B	598.4	[579.5, 623.1]	8.9%

Table 4: Validation HumorRank Leaderboard (Qwen 2.5 72B Judge). The hierarchy maintains strong alignment with the primary leaderboard ($\tau = 0.889$), cleanly separating the frontier models, the mid-tier models (including the domain-specialist HumorGen 7B), and the weak baselines.

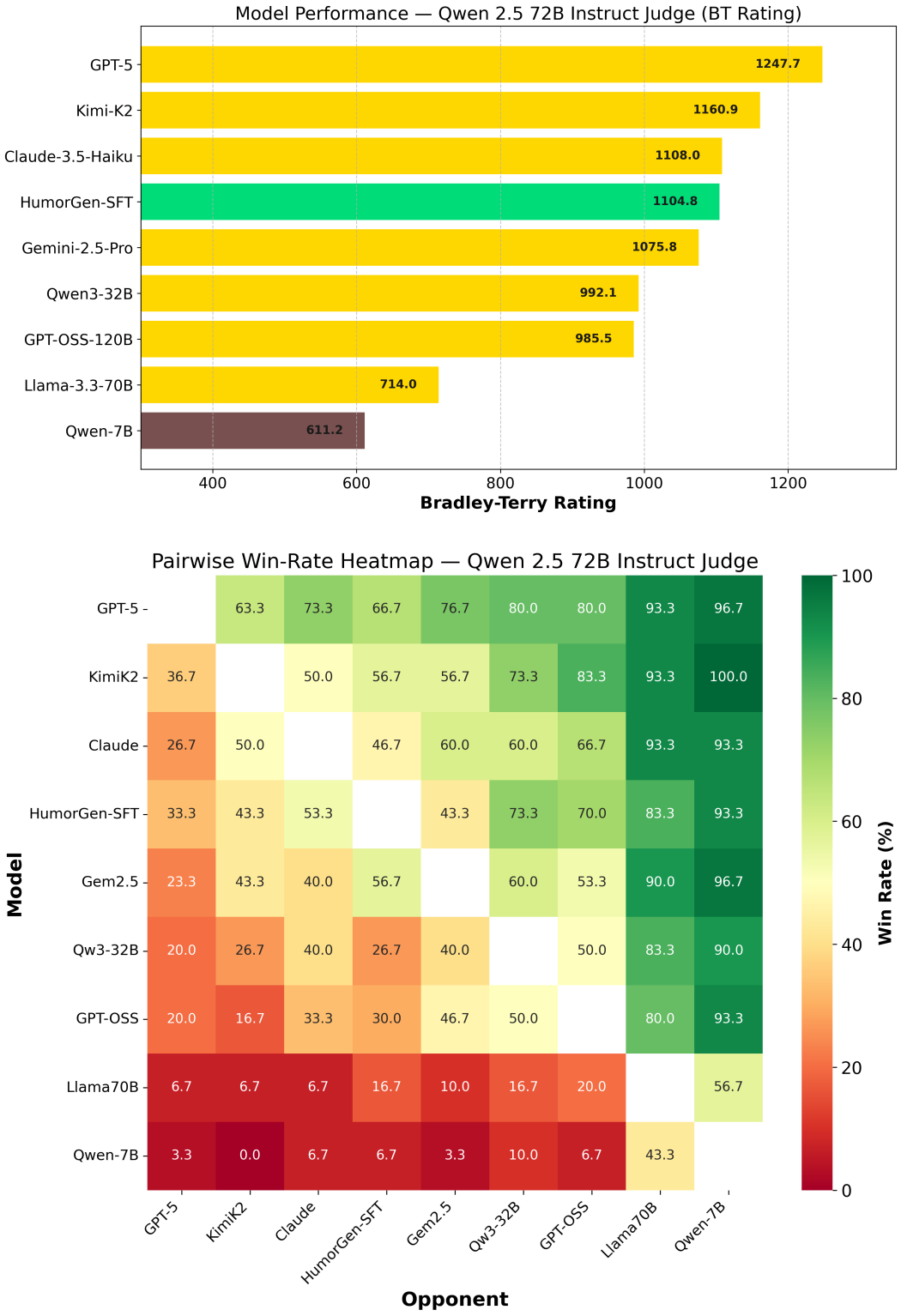


Figure 4: HumorRank Leaderboard (top) and Pairwise Win-Rate Heatmap (bottom) showing the performance of the 9 models.

C LLaMA Judge Sample Decision Making

Below are representative LLaMA judge decisions drawn from the evaluation log. Each pair shows the preferred joke (green) against the rejected entry (red), together with the judge's reasoning and ELO updates.

Winning Features Delivery Features Loser Features

Decision #1 en_2051 · Tie: No · Confidence: MEDIUM · Score: 1.0 · ELO_Δ: +7.66 / −7.66

Headline: 'We did it!': Countries celebrate World Cup qualification

Reasoning: Joke A has a more relatable and unexpected twist—the concept of "not washing lucky socks" as a claimed contribution to the team's success makes it more humorous and engaging.

Winning Features: incongruity · absurdity · surprise · irony

Delivery Features: conciseness · punchline_positioning · framing_commitment

Loser Features: cliché · weak_punchline

Gemini-2.5-Pro · Winner

After my team qualified for the World Cup, I ran into the street yelling, "WE DID IT!" My neighbor looked at me and said, "What exactly did *you* do?" I told him, "I successfully avoided jinxing them by not washing my lucky socks for six months. You're welcome."

Claude-3.5-Haiku · Loser ✗

Why did the soccer player bring a ladder to celebrate World Cup qualification? Because they wanted to take their excitement to the next level... literally, they were already on cloud nine!

Decision #2 en_2021 · Tie: No · Confidence: MEDIUM · Score: 1.0 · ELO_Δ: +7.14 / −7.14

Headline: 'Mind-blowing' population of feral goldfish in Ottawa pond to be euthanized

Reasoning: Joke A uses a clever play on words with 'mind-blowing' and creates a vivid, absurd image; Joke B relies on a more straightforward and less surprising punchline.

Winning Features: wordplay · absurdity · incongruity

Delivery Features: conciseness · punchline_positioning · framing_commitment

Loser Features: cliché · weak_punchline

HumorGen-SFT-7B · Winner

So, they're going to euthanize these 'mind-blowing' feral goldfish in Ottawa's pond? I hope they have anesthetic for fish because if you don't do it right, those goldfish might just get their last swim as they float up to heaven saying, "Mind-blowing, indeed!"

GPT-OSS-120B · Loser ✗

Ottawa's pond got so many feral goldfish that the city had to "euthanize" them—turns out they were just trying to pay the water bill with their own scales.

Decision #3 en_2034 · Tie: No · Confidence: MEDIUM · Score: 1.0 · ELO_Δ: +7.63 / −7.63

Headline: Three-quarters of parents let children miss school for 'duvet day'

Reasoning: Joke B has a more complex and layered structure—the fraction wordplay ("three-quarters is exactly the blanket-to-face ratio") adds an extra dimension of sarcasm and irony over Joke A's simpler setup.

Winning Features: incongruity · wordplay · absurdity · irony · sarcasm

Delivery Features: timing · conciseness · punchline_positioning · framing_commitment

Loser Features: weak_punchline

GPT-5 · Winner

Three-quarters of parents let kids miss school for "duvet day"—teachers are fuming: "How are we supposed to cover the syllabus when everyone's already under the covers?" On the bright side, kids finally understand fractions: three-quarters is exactly the blanket-to-face ratio required to breathe.

Base-Qwen-7B · Loser ×

Why did the kid stay home from school? Because when asked if he wanted to go, he replied, "I'd rather be sick three-quarters of the time!"

Decision #4 · en_2115 · Tie: No · Confidence: MEDIUM · Score: 1.0 · ELO_Δ: +7.66 / −7.66

Headline: How to design an aesthetic home office that boosts productivity

Reasoning: Both jokes rely on irony and sarcasm, but Joke A's darker, more unexpected twist—"the exact shade of my laptop screen at 3 a.m.—black with a faint glow of regret"—is more surprising and absurd.

Winning Features: incongruity · absurdity · surprise · irony · sarcasm

Delivery Features: conciseness · deadpan · punchline_positioning

Loser Features: cliché · weak_punchline

Kimi-K2 · Winner

My productivity skyrocketed after I designed my aesthetic home office: I painted the walls the exact shade of my laptop screen at 3 a.m.—black with a faint glow of regret.

Gemini-2.5-Pro · Loser ×

I spent all weekend creating the perfect aesthetic home office to boost my productivity. It worked! I am now incredibly productive at taking pictures of my aesthetic home office for Instagram.

Figure 5: Four representative LLaMA judge decisions. Winner (green) and Loser × (red) are labelled directly on each joke box. Feature rows indicate winning humor traits (green), delivery strengths (blue), and loser weaknesses (red). ELO deltas are approximated from the evaluation log.

D Hyperparameter Configurations

Standardized hyperparameters across the HumorRank tournament are detailed in Table 5 (candidate generation) and Table 6 (LLM adjudication). All tournament-level Bradley–Terry and Elo parameters are summarized in Table 7.

Parameter	Value
Temperature	0.7
Top-p	0.9 (Local Models) / 1.0 (API Default)
Max New Tokens	256
System Prompt	"You are a joke generator. Given a headline or topic, generate a funny joke. Output ONLY the joke text. No thinking tags, no reasoning, no explanation, no extra words."

Table 5: Hyperparameters for candidate humor generation across all local and API-based models. API-based models used default top-p values if not explicitly exposed.

Parameter	Value
Primary Judge	Llama 3.3 70B Instruct
Ablation Judge	Qwen 2.5 72B Instruct
Temperature	0.1
Max New Tokens	512
Max Retries	3
Backoff Base	2.0
Retry Cap	4.0

Table 6: Hyperparameters for LLM-as-a-Judge adjudication.

Parameter	Value
Initial Elo Rating	1000.0
K-factor	32
Min Rounds per Model	2
Max Rounds per Model	3
Stable Elo Shuffles	10
BT Convergence (ϵ)	10^{-6}
Bootstrap Iterations	100

Table 7: Tournament configuration and Bradley–Terry global MLE parameters.

E LLM-as-a-Judge Prompting Framework

The following prompt template was used for all pairwise comparisons in the HumorRank evaluation pipeline. The judge models received a system prompt establishing their role as comedy critics, followed by a structured user prompt presenting two jokes for comparison. All 10,800 automated tournament comparisons for each judge utilized this exact template.

system PROMPT_V3 — System

```
"You are a comedy critic judging which of two jokes is funnier.\n"
"Analyze both the underlying logic (humor mechanisms) and the presentation (delivery).\n"
"Be direct and honest. If one joke is clearly better, pick it. "
"If they are genuinely equal in quality, say TIE.\n"
"Do not overthink it --- trust your first impression. Output JSON only."
```

user PROMPT_V3 — User

```
'Prompt: "{headline}"\n\n'
"JOKE A: {joke_a}\n\n"
"JOKE B: {joke_b}\n\n"
"Which is funnier? Return JSON:\n"
"{{\n"
'  "reasoning": "brief explanation",\n'
'  "decision": "A" or "B" or "TIE",\n'
'  "winner_humor_features": [list ALL that apply, 1-3, from: {mech_features}],\n'
'  "winner_delivery_features": [list ALL that apply, 1-3, from: {deliv_features}],\n'
'  "loser_features": [list ALL that apply, 1-3, from: {loser_features}]\n'
"}}"
```

HUMOR MECHANISMS

"incongruity", "wordplay", "absurdity", "surprise", "irony", "sarcasm", "observational", "narrative"

DELIVERY FEATURES

"timing", "conciseness", "deadpan", "escalation", "punchline_positioning", "framing_commitment"

LOSER FEATURES

"cliché", "confusing", "offensive", "overexplained", "buried_punchline", "weak_punchline"

Figure 6: Prompt template used for all 10,800 pairwise comparisons. Template variables ({headline}, {joke_a}, {joke_b}) are instantiated per comparison. The three feature lists (humor mechanisms, delivery, and loser features) enforce structured and consistent JSON outputs across all evaluations.

F Qualitative Examples and Feature Reasoning

A representative pairwise qualitative example is presented in the main paper (Table 3). The per-model feature distributions adjudicated by the primary Llama 3.3 70B judge are shown in Figure 2 in the main paper. The Qwen 2.5 72B distributions are reproduced below (Figure 7) for cross-judge comparison.

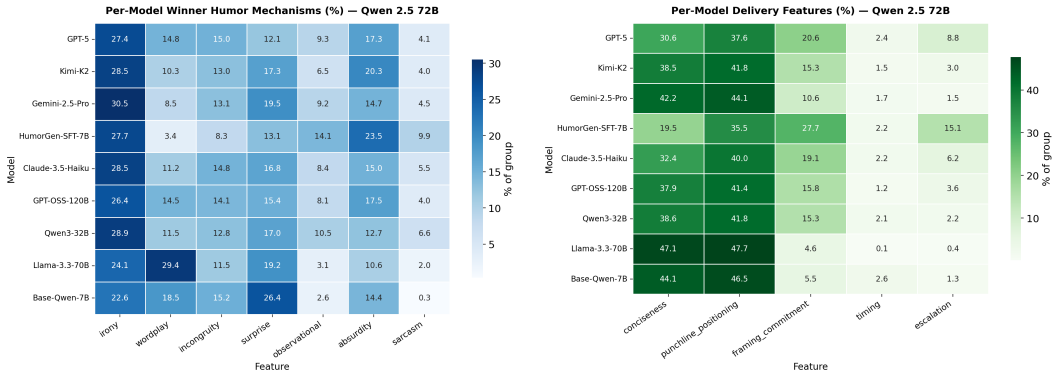


Figure 7: Per-model winning feature distributions (Qwen 2.5 72B judge). **Left:** Humor mechanisms. **Right:** Delivery features. Rank patterns are consistent with the primary Llama judge (Figure 2), confirming cross-judge stability.

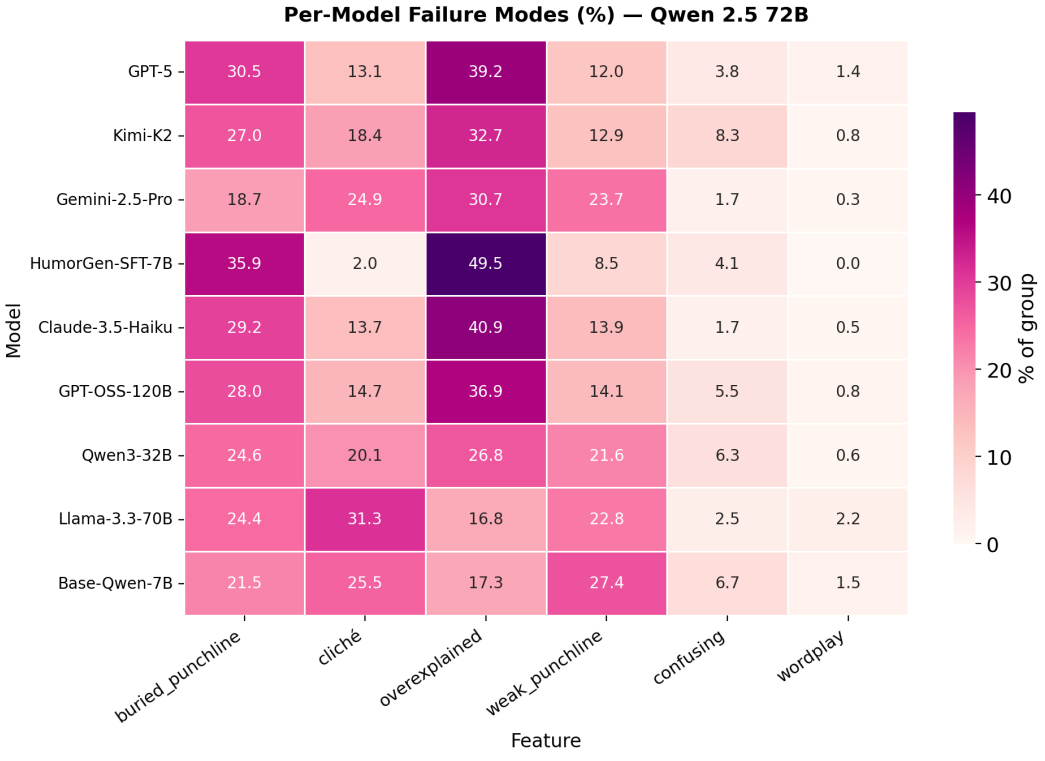


Figure 8: Per-model failure mode distributions (Qwen 2.5 72B judge). HumorGen SFT 7B again shows markedly higher *Overexplained* (49.5%) rates compared to other models, consistent with findings under the primary Llama judge (Figure 3), confirming that this failure signature is evaluator-agnostic.

F.1 Key Observations & Findings

A comprehensive analysis of the heatmaps across both judge paradigms reveals several profound insights regarding how different model architectures approach humor generation:

1. **The Shallow Heuristics of Open-Weight Baseline Models:** As visualized across both evaluations, the baseline open-weight models (e.g., Base-Qwen-7B and Llama-3.3-70B) demonstrate a heavy reliance on shallow lexical heuristics. In terms of mechanics (Figures 2 and 7, Left), they rely almost exclusively on superficial *Wordplay* (at nearly 30% of their wins). In delivery (Figures 2 and 7, Right), this heuristic behavior is even more extreme: these models devote over 85% of their winning generations to simple *Conciseness* and strict *Punchline Positioning*, effectively functioning as rigid, template-based joke outputters.
2. **The Structural Depth of Specialized Alignment:** Conversely, the fine-tuned specialist (HumorGen-SFT-7B) exhibits a radically different and deeper psychometric signature. For mechanics (Figures 2 and 7, Left), it outputs the highest rates of *Absurdity* (over 23% under both judges) and *Sarcasm*, successfully wielding semantic incongruity over cheap wordplay. In delivery (Figures 2 and 7, Right), rather than dumping immediate punchlines, HumorGen uniquely leverages *Framing Commitment* (approx. 27%) and *Escalation* (approx. 15%), proving it can hold the setup and build narrative tension.
3. **The Balanced Proprietary Frontier:** Frontier models like GPT-5 and Kimi-K2 sit carefully between these extremes. While their delivery aesthetics are generally safe—indexing highly on conciseness and optimal positioning (Figures 2 and 7, Right)—they maintain enough cognitive depth to succeed heavily on *Irony* and *Incongruity* without having to resort exclusively to the aggressive absurdity of the specialist or the simplistic puns of the baselines.

G Human Evaluation Protocol and Annotator Guidelines

The human evaluation protocol is designed to minimize bias and ensure high-fidelity preference judgments. Annotators are presented with the original setup and two model-generated responses (labeled anonymously as 'Joke A' and 'Joke B'). They are instructed to evaluate which response is funnier based on the context, or formally declare a tie if both are equally humorous or equally poor.

Inter-Annotator Reliability (Krippendorff's Alpha) Due to the highly subjective nature of humor preference and the potential for sparse pairwise intersections, we utilize Krippendorff's Alpha (α) to rigorously quantify inter-annotator agreement. The metric is defined as:

$$\alpha = 1 - \frac{D_o}{D_e} \tag{3}$$

where D_o represents the observed disagreement among annotators, and D_e represents the disagreement expected by chance. For our pairwise preference schema (Model A vs. Model B vs. Tie), we compute α using a nominal distance metric. An $\alpha > 0$ indicates agreement beyond chance; notably, values approaching 0.45 traditionally indicate moderate agreement, which is standard for highly subjective NLP generation benchmarks such as computational comedy evaluation.

Our human evaluation yields a raw human agreement of 52.6%, an $\alpha = 0.446$, and a 49.0% micro-averaged alignment with the primary LLM judge. Complete cohort demographics and comprehensive statistical alignment tables will be formally incorporated upon final publication.

H Human Evaluation Details

Instructions to participants. Evaluators were shown the following instructions before starting (see Figure 9 for the screen as displayed).

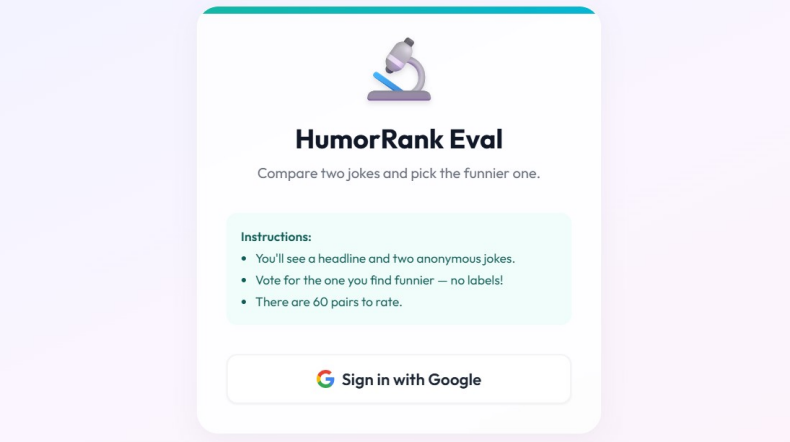


Figure 9: Instructions screen (HumorRank Blind Evaluation) As shown to participants before voting.

The human evaluation protocol is designed to minimize bias and ensure high-fidelity preference judgments. Annotators are presented with the original setup and two model-generated responses (labeled anonymously as 'Joke A' and 'Joke B'). They are instructed to evaluate which response is funnier based on the context, or formally declare a tie if both are equally humorous or equally poor.

Metrics and recruitment. Our evaluation comprised 120 total votes (2 expert evaluators, 60 pairs). Evaluators were recruited by invitation; no payment was provided. The final study yields a Human agreement of 50.0%; an LLM vs. consensus (Gold Standard) alignment of 53.3%; and a micro-average accuracy of 51.7%. Position bias was mitigated via random A/B assignment.

Agreement definitions. We report the following three agreement metrics:

1. **Human agreement** (inter-annotator): The proportion of pairs for which all annotators selected the same winner. Formally, the number of pairs with unanimous agreement divided by the total number of pairs.
2. **Gold Standard agreement** (LLM–consensus): The proportion of pairs in which the LLM judge's choice coincides with the majority vote among human annotators for that pair. Computed as the number of pairs where the LLM prediction matches the human consensus, divided by the total number of pairs.
3. **Micro-average accuracy**: The proportion of all individual human votes (across evaluators and pairs) that agree with the LLM's choice. Computed as the number of votes matching the LLM divided by the total number of votes.

Inter-Annotator Reliability (Krippendorff's Alpha) Due to the highly subjective nature of humor preference and the potential for sparse pairwise intersections, we utilize Krippendorff's Alpha (α) to rigorously quantify inter-annotator agreement. The metric is defined as:

$$\alpha = 1 - \frac{D_o}{D_e} \tag{4}$$

where D_o represents the observed disagreement among annotators, and D_e represents the disagreement expected by chance. For our pairwise preference schema (Model A vs. Model B vs. Tie), we compute α using a nominal distance metric. Our evaluation yields an $\alpha = 0.432$, which traditionally indicates moderate agreement and is standard for highly subjective NLP generation benchmarks such as computational comedy evaluation.

I Mathematical Stability of the Generative Tournament

To validate the robustness of the derived Elo ratings against sequence-dependence (often referred to as "late-winner bias" in streamed continuous tournaments), HumorRank utilizes a Stable Elo variant grounded in order-independent aggregation (Albers & Vries, 2001).
In standard sequential Elo implementations, a model m updates its rating R_m after a sequence of matches based on the standard iterative update rule:

$$R_m^{(t+1)} = R_m^{(t)} + K \cdot (S - E_m) \tag{5}$$

where t indexes the chronological order of the match. Consequently, a model earning a win at the end of the match history block gains an inherently outsized advantage over a model that earned an identical win early in the sequence.
To completely nullify this temporal artifact, we strip the time dependencies by evaluating the match history H across N independently shuffled topological permutations. The stable terminal rating $\bar{R}_m$ for each model m is defined as the arithmetic mean across all sequences:

$$\bar{R}_m = \frac{1}{N} \sum_{k=1}^{N} R_{m,k}^{(T)} \tag{6}$$

where $R_{m,k}^{(T)}$ represents the final rating of model m after iterating through all $T = 10,800$ matches in the k-th shuffled permutation.
In our experiments, we set $N = 10$ to ensure convergence. To empirically validate that this methodology produces a rigid transitive hierarchy rather than high-variance noise, we measured the standard deviation of final ratings across the permutations:

$$\sigma_m = \sqrt{\frac{1}{N} \sum_{k=1}^{N} \left(R_{m,k}^{(T)} - \bar{R}_m \right)^2} \tag{7}$$

Tracking this distribution across both the primary (Llama 3.3 70B) and validation (Qwen 2.5 72B) judges for all 9 contestants yielded the following internal stability metrics on a base 1000-point scale:

- **Maximum Variance:** Bounded strictly at $\sigma_{max} = 37.5$ Elo points across all models.
- **Mean Variance:** Clustered tightly around $\bar{\sigma} \approx 29.5$ Elo points.

Given that inter-model spreads on the leaderboard exceed 200 points, this stringent empirical convergence ($\sigma < 38.0$) confirms that the generative tournament produced a highly stable latent hierarchy entirely free from tournament-ordering artifacts.